Executive Summary: Detecting Frozen & Stuck Waste Left Behind After Pickup

- **The Problem (What Fill Sensors Miss in Winter):** In Nordic winters, wet biowaste and high-moisture mixed waste freeze to the floor and walls of **COMBI** containers (`<= 22 m³`). When the collection truck tips the container and returns it to the bay, the hydraulic press plate retracts to its home position – causing existing **Europress SMART** fill sensors to report the container as empty (`0%–13.7%` fill) even when **300 to 700 kg of frozen waste is still stuck inside**.

- **Why Customers Lose Money:** Without weight data, customers pay **100% of the fixed lift and transport fee (€110–€350)** for a container that came back nearly a third full by weight, lose usable capacity on the very next cycle, and risk miscalculated net collection weights.

- **How Our Solution Works Inside Europress SMART:**

  - **Post-Lift Residual Weight Check:** Immediately after each pickup, SMART compares post-lift residual weight (W_{after}) against pre-lift container weight (W_{before}) alongside existing hydraulic fill data.

  - **Neutral Calibration Guardrail:** Before alerting on stuck waste, SMART cross-checks maintenance logs and multi-pickup baseline offsets to separate **sensor zero-offset drift** (such as a `~196 kg` load-cell offset) from physical waste – **never blaming the waste operator** for sensor drift and instead scheduling a **Europress technician calibration visit (€185/visit)**.

  - **Actionable Winter Upsell:** When sensors are healthy, SMART flags genuine stuck waste, records true net collected weight, and recommends a retrofittable **heated container floor (+€35/device/month)** on winter **COMBI** units.

- **Measurable Value & Business Case:** Identifies **180 winter partial-empty pickups** (`312 kg` / `27.3%` median stuck waste at `13.7%` fill) and **2,083 pickups** with `>= 80 kg` residual weight in the competition dataset (`16` pickups in quickstart: `2` physical stuck loads vs. `14` sensor-offset pickups), anchoring a **€29–€49/device/month** SMART SaaS tier with a **1.6-month customer payback** (`2.4×–4.6×` ROI) and **€463,400/yr** net customer benefit across the full weight suite.

Detecting Frozen & Stuck Waste Left Behind After Pickup

Why It Matters (What Fill Alone Cannot See)

In Nordic winters, wet biowaste and high-moisture mixed waste freeze to the floor and walls of **COMBI** containers (`<= 22 m³`). When the collection truck tips the container and returns it to the bay, the compactor's hydraulic press plate is retracted to its home position. Because existing **Europress SMART** fill measurement relies on ram stroke and hydraulic pressure, the post-lift fill level drops near zero (`0%–13.7%`) even when **300 to 700 kg of waste is still stuck inside the container**.

Without weight measurement, the fill sensor reports the container as empty, so the customer pays **100% of the fixed lift and transport fee (€110–€350)** for a container that came back nearly a third full by weight, their net collected weight is miscalculated unless corrected, and their next collection cycle fills up days earlier than expected.

How It Works (Fusing Weight with Existing Fill & Maintenance Data)

- **Step 1 – Post-Lift Residual Weight & Share Calculation:** Immediately after pickup completion, SMART reads the **post-lift residual weight** (W_{after}) from the compactor's load cells and compares it against the **pre-lift container weight** (W_{before}):

$$\text{Residual Share} = \frac{W_{\text{after}}}{W_{\text{before}}}$$

- **Step 2 – Neutral Sensor Calibration Guardrail:** Before flagging a pickup as a partial empty, our engine cross-checks **maintenance and calibration records** (such as sensor recalibration or load-cell replacement logs) and multi-pickup baseline offsets. If a compactor exhibits a constant baseline offset across consecutive pickups followed by a technician calibration visit (for example, a unit showing a `~196 kg` sensor zero-offset resolved by a calibration work order), the system classifies it as an **Equipment Calibration Offset** and **never blames the waste operator**.
- **Step 3 – Stuck-Waste Alert & Winter Equipment Recommendation:** Only when the sensor baseline is normal does SMART flag genuine **physical residual waste stuck inside the container**, automatically protecting the customer's true net collected weight (pre-lift minus post-lift weight) and recommending a retrofittable **heated container floor (+€35/device/month)** on winter **COMBI** biowaste units.

Measurable Customer Benefit & Dataset Evidence

- **Full Competition Dataset (447 public devices, 365 days):** Identifies **180 partial-empty pickups** where a median of **312 kg** (`27.3%` of the pre-lift load, up to `710 kg`) remained stuck inside the container while the post-lift fill sensor read only **13.7%**, plus **2,083 completed pickups** with `>= 80 kg` post-lift residual weight at `2.9%` median fill (`Estimated` post-lift load-cell weight + `Observed` maintenance logs).
- **Quickstart Dataset (50 devices, 90 days):** Identifies **16 pickups** with `>= 80 kg` post-lift residual weight – correctly separating **2 genuine physical partial empties** from **14 pickups caused by a ~196 kg sensor zero-offset** resolved by a calibration work order.
- **Direct Value to Customer & Europress:** Protects customers from paying full lift fees for unemptied winter containers, ensures accurate net collected weight on every pickup, and gives Europress a high-conversion hardware upsell (**retrofittable heated container floors at +€35/device/month**) alongside targeted field calibration visits (`€185/visit`).
- **Additional Fleet-Wide Value from Supporting Capabilities (Appendix Ideas 1–6):** When combined with the six additional weight capabilities in the Appendix – detecting recyclables thrown into mixed waste (`17` multi-stream compactors with `588.8 t/yr` of divertable recyclables = `€108,930/yr`), fair weight splitting on 2-stop shared truck routes (`532` combined-load pickups), verifying compaction force and upgrading overworked COMBI units to DUO (`38` high-frequency units = `+€50,160/yr` lease upgrades), separating truck stored-tare errors from sensor drift (`18` biased trucks across `1,531` pickups = `€12,206` vs. `63` sensor drift episodes), instant pickup weight receipts & unlocking recyclable credits (`4,908` unshared pickups + `€86,923` single-load billing gaps + `€365,671/yr` across `36` zero-credit recyclable contracts), and ordering pickups at target weight to avoid

minimum-charge penalties (`2,560` penalty pickups = `€192,789.15` observed penalties / `€303,040/yr` counterfactual dispatch savings) – the full SMART Weight suite delivers **€463,400/year in net customer benefit** in our recommended base case.

COMMERCIAL MODEL FOR EUROPRESS & PRICING JUSTIFICATION

Three Complementary Revenue Streams for Europress

- **1. Tier 1 – SMART Weight Audit & Receipt (€29 / device / month or €348/yr):**

Designed for single-stream weighed compactors (**COMBI** or **DUO**). Includes our core **Detecting Frozen & Stuck Waste Left Behind After Pickup** capability, plus Day-0 `±3.3%` weight receipts and separating truck tare errors from compactor sensor drift (Appendix Ideas 4 and 5). An average single-stream weighed compactor in the dataset shows **€1,320/yr** in identifiable value, giving the customer a **3.8× gross value-to-fee ratio** (~26% value capture by Europress).

- **2. Tier 2 – SMART Weight Tonnage & Sorting Optimizer (€49 / device / month or €588/yr):**

Designed for multi-stream sites and shared-load routes. Includes all of Tier 1 plus detecting recyclables thrown into mixed waste (`€6,408/yr` average sorting value per affected mixed-waste unit), fair weight splitting on 2-stop shared truck routes, and target-weight minimum-charge dispatch guards (Appendix Ideas 1, 2, and 6). Multi-stream and minimum-charge-exposed compactors show **€2,710/yr** in average gross value – a **4.6× gross value-to-fee ratio** (~22% value capture by Europress).

- **3. Tier 3 – Hardware Lease & Field Service Pull-Through:**

Turns weight telemetry directly into core Europress machine and service sales:

  - **Winter Heated Container Floors (+€35/month):** Stuck frozen load alerts directly drive retrofittable heated floor add-ons on winter biowaste **COMBI** units.
  - **Targeted Field Calibration Visits (€185/visit):** Billable technician calibration dispatches triggered when baseline offset checks (in our core frozen-waste detection and Appendix Idea 4) isolate compactor sensor drift.
  - **Dedicated Recyclable Balers & Compactors:** Mixed-waste density alerts (Appendix Idea 1) act as a qualified lead-generation engine to lease additional **Europress Balex** balers and dedicated **DUO** recyclable compactors to sites losing `€6,000+/yr` on poor sorting.
  - **COMBI-to-DUO Lease Upgrades (+€110/month or +€1,320/yr incremental lease):** Upgrading high-frequency non-biowaste **COMBI** units (`60+ lifts/yr`) to larger `24–30 m³` **DUO** units saves the customer `€2,500–€4,200/yr` in lift fees (**2.4× customer ROI**) while generating **+€50,160/yr** in recurring lease upgrades across `38` candidate units (Appendix Idea 3).

- **Cost to Serve & Margin (€8.50 / device / month or €102/yr):**

Covers cellular telemetry overhead, cloud ingestion, and automated calibration monitoring – yielding an **~81% SaaS gross margin** for Europress.

- **Sensitivity Across Five Scenarios (52 Customers, 600 Devices, 464 Weighed):**
  - **Base Case (Recommended – 42% adoption, 65% realisation, ±3.3% sensor accuracy):** Generates **€463,400/yr** in customer net benefit, **€145,600/yr** in Europress ARR (`€124,200/yr` net margin after cost to serve), and a **1.6-month customer payback**.
  - **Half Adoption Stress Test (21% adoption, 65% realisation):** Generates **€231,700/yr** in customer net benefit, **€72,800/yr** in Europress ARR (`€62,100/yr` net margin), and a **1.6-month payback**.
  - **Conservative Realisation (42% adoption, 32.5% realisation):** Generates **€189,800/yr** in customer net benefit, **€145,600/yr** in Europress ARR, and a **3.3-month payback**.
  - **Degraded Weight Sensor (35% adoption, 50% realisation, ±6.5% sensor noise):** Generates **€241,500/yr** in customer net benefit, **€121,300/yr** in Europress ARR, and a **2.2-month payback**.

- **Combined Worst Case (20% adoption, 35% realisation, ±6.5% sensor noise):** Customers still remain net-positive (**€86,200/yr net benefit**, **3.8-month payback**) and Europress earns **€46,100/yr net contribution margin** (`79%` margin).

PRODUCTISATION AS PART OF EUROPRESS SMART

Rather than requiring a separate standalone platform, our solution is designed to plug directly into the four core service pillars that **Europress SMART** already provides today – **equipment & fill-level monitoring**, **collection ordering**, **maintenance information**, and **customer reporting** – by fusing the new integrated weight signal with the volume and machine data Europress already collects:

- **1. Fusing Weight with Existing Volume & Fill-Level Monitoring (kg + m³ + Fill):**
  - Europress SMART already knows each machine's container volume (`<= 22 m³` for **COMBI**, `<= 30 m³` for **DUO**), hydraulic press cycles, and pressure-based fill level over its existing cellular telemetry link.
  - By combining live container weight with existing fill-level and volume data on the same SMART device stream, SMART unlocks **post-lift residual weight** and **compacted bulk density (kg/m³)** with zero additional site hardware or optical cameras.
  - This fused weight-plus-volume layer directly powers **Detecting Frozen & Stuck Waste Left Behind After Pickup** (detecting when a container returns from emptying with near-zero hydraulic fill but hundreds of kilograms of frozen waste still stuck inside), as well as detecting recyclables thrown into mixed waste (Appendix Idea 1) and verifying compaction pressure adjustments (Appendix Idea 3).
- **2. Upgrading SMART Collection Ordering from Fill-Only to Weight-Aware Dispatch:**
  - Today, SMART triggers collection orders when a compactor's pressure-based fill estimate crosses a volume threshold or when site staff press a manual order button.
  - Integrating weight into SMART's existing collection-ordering workflow adds a **Target-Weight Dispatch Guard** (Appendix Idea 6): on per-tonne and hybrid contracts, SMART checks both volume fill and live container weight against the contract's minimum-charge break-even tonnage – holding premature or precautionary Friday pickups until the container reaches economic payload weight, while always enforcing the 7-day biowaste hygiene limit and zero-overflow protection.
- **3. Linking Weight Telemetry to SMART Maintenance & Field Service:**
  - Europress SMART already tracks equipment status, alarms, and technician service history.
  - Connecting weight readings directly to SMART's maintenance workflow (in our core frozen-waste detection and Appendix Idea 4) cross-checks multi-pickup baseline offsets and compares compactors against live-weighed collection trucks across the fleet – automatically separating **equipment sensor drift or zero-offset** from **physical stuck waste** or **hauler stored-tare errors**, suppressing false operator alerts, and scheduling a targeted **Europress technician calibration visit**.
- **4. Extending SMART Customer Reporting with Contract Verification & Human-Approved Decisions:**
  - Within SMART's existing reporting portal, customers can optionally attach their operator contract rates, monthly invoices, and weighbridge tickets.
  - SMART automatically logs **post-lift residual weight alerts**, quantifies **recyclable sorting savings** (Appendix Idea 1), issues **Day-0 Weight Receipts** after every pickup (Appendix Idea 5), proportionally reconciles **50/50 combined-load tickets** across paired sites (Appendix Idea 2), and surfaces neutral, evidence-backed recommendations behind **explicit human review and approval**, with every figure tagged as `Observed`, `Estimated`, or `Hypothetical`.

ASSUMPTIONS & WHAT REMAINS UNTESTED

Assumptions Made

- **Weight Source Priority Chain:** Wherever contract billing or cost per tonne is computed, our engine strictly follows `Certified Weighbridge Ticket -> Billed Invoice Weight -> Calibrated Compactor Weight -> Unknown (NULL)`. Unknown weights are **never imputed or guessed**.
- **Time Zones & Contract Selection:** All timestamps are stored in UTC and converted to **Europe/Helsinki** local dates to select the active customer contract version and monthly recyclable material price index.
- **Operational Tolerances:** Invoice line rounding follows half-away-from-zero to the integer cent; invoice-vs-contract line tolerance is `2%`; single-load compactor-vs-ticket discrepancy review threshold is `10%` (and `±3.3%` under the hypothetical Day-0 weight verification clause); minimum sample size for high-confidence operator scorecard judgement is `>= 10` pickups.
- **Commercial Parameters:** Monthly SaaS fees (`€29/mo` Tier 1, `€49/mo` Tier 2), hardware add-on lease deltas (`+€110/mo` **DUO** upgrade, `+€35/mo` heated biowaste floor, `€185` calibration visit), cost to serve (`€8.50/device/month`), base adoption (`42%`), and base human-approved realisation rate (`65%`) are explicit team assumptions grounded in dataset value ratios (`2.4×–4.6×` customer ROI).
- **Credit-Eligible Materials & Equipment Limits:** Material credits apply strictly to **cardboard**, **plastic film**, and **mixed plastic** (paper, mixed waste, and biowaste are always chargeable). Equipment classes strictly respect **COMBI** `<= 22 m³` (including all biowaste) and **DUO** `<= 30 m³` (including all mixed plastic).

What Remains Untested & Unproven in the Real World

- **No Physical Weighing Prototype (Hardware Accuracy & Survivability Untested):**

The dataset assumes an integrated weighing accuracy of `±3.3%` (`~100 kg` on a `3 t` load) in normal operation. In the real world, achieving `±3.3%` on a **COMBI** hook-lift subframe or a **DUO** detachable container under `200+ bar` hydraulic compaction forces, uneven asphalt/snow inclination (`1°–3°`), and `-25 °C` Finnish winter freeze-thaw cycles is **untested in this software submission** and requires physical pilot validation at Europress's Kerava facility.

- **Synthetic Data & Counterfactuals Are Not Market Proof:**

All waste generation rates, compacted densities (`380 kg/m³` mixed waste vs. `190 kg/m³` cardboard), operator sharing lags (`0–45` days), invoice error rates (`1.0–3.5%`), and counterfactual dispatch outcomes come from the synthetic simulator. They demonstrate that our reasoning and software work correctly, **not** that real customers will achieve these exact euro savings.

- **Commercial Willingness-to-Pay (WTP) & Operator Contract Acceptance:**

While the dataset shows strong gross value per device, real-world customer willingness to pay `€29–€49/device/month`, operator willingness to accept source-weight verification clauses or proportional combined-load splits, and legal metrology requirements (`NAWI Directive 2014/31/EU` Class III/IIII for direct billing) remain **untested commercial hypotheses**.

- **Negotiation & Tender Dynamics:**

Expected values for invoice disputes, weighbridge data requests, contract renegotiations, and operator tenders rely on benchmark recovery probabilities and effort costs. Real-world waste contract negotiations depend on multi-site bundling, regional competition, municipal regulations, and long-term commercial relationships not captured in a 90-day or 365-day window.

APPENDIX: OTHER IDEAS FOR USING WEIGHT DATA

Alongside **Detecting Frozen & Stuck Waste Left Behind After Pickup**, integrating weight into **Europress SMART** unlocks six additional high-value capabilities across multi-stream sorting, shared collection routes, fleet right-sizing, sensor maintenance, contract verification, and dispatch optimisation:

Appendix Idea 1: Detecting Recyclables Thrown into Mixed Waste

- **Why It Matters:** Material quality is determined only by the waste operator at its receiving facility, and Europress has no optical camera technology inside the container to inspect material quality before collection. On a mixed-waste compactor, a fill sensor only sees that the container is `80%` full. However, compacted **mixed waste** has a nominal bulk density of **380 kg/m³**, whereas **cardboard (190 kg/m³)**, **plastic film (140 kg/m³)**, and **mixed plastic (115 kg/m³)** have less than half that density (`~205 kg/m³` blended). When retail or warehouse staff throw uncrushed cardboard boxes and plastic wrap into the general mixed-waste compactor, its occupied volume fills up rapidly while its weight stays abnormally low.
- **How It Works:**
  - For every pickup with a valid pre-lift fill reading (`>= 30%`) and calibrated weight, SMART fuses its existing container volume ($V_{\text{container}}$) and pre-lift fill level ($Fill_{\text{before}}$) with the net collected weight ($W_{\text{collected}}$) to calculate the **effective compacted bulk density** ($rho_{\text{compacted}}$):

$$Density_{\text{compacted}} = \frac{W_{\text{collected}}}{V_{\text{container}} \times Fill_{\text{before}}}$$

  - When a mixed-waste compactor chronically operates well below `380 kg/m³` (for example, operating at `247.1 kg/m³` or `255.8 kg/m³`), a two-component physical density balance estimates the weight fraction ($alpha_{\text{recyclable}}$) of low-density recyclables mixed into the general waste stream:

$$Share_{\text{recyclable}} = \text{clip}\left(\frac{380 - Density_{\text{compacted}}}{380 - 205},\ 0\%,\ 65\%\right)$$

- **Value to Customer & Europress:** Every tonne of cardboard or plastic thrown into mixed waste costs the customer `~€120/t` in mixed-waste processing fees *instead* of earning `~€65/t` in recyclable material credits – a **€185/t net financial swing** (`€108,930/yr` across `588.8 t/yr` on `17` multi-stream mixed-waste compactors in the competition dataset, and `€10,285/yr` across `55.6 t/yr` on `7` devices in the quickstart dataset, tagged `Hypothetical`). For Europress, this turns weight telemetry into a **lead-generation engine to sell additional Europress Balex balers and dedicated DUO recyclable compactors** at the exact sites losing money on poor sorting.

Appendix Idea 2: Fair Weight Splitting When One Truck Collects Two Containers

- **Why It Matters:** To reduce transport emissions and route costs, haulers frequently collect two containers on a single truck trip (`532` combined-load pickups across `266` paired trips in the competition dataset). Because the receiving facility weighbridge only weighs the truck once upon arrival, the operator allocates the certified scale weight **50/50** between the two pickups. Fill level cannot challenge a 50/50 split because two containers may have different volumes (`10 m³` vs. `24 m³`) or different compacted densities (resulting in a `450 kg` median allocation error and `63.6%` of combined pickups erring by `>15%`).
- **How It Works:**
  - SMART pairs combined-load pickups that share the same weighbridge ticket where both compactors have calibrated weight measurements (m_A and m_B).
  - Instead of an arbitrary 50/50 split, SMART reconciles the operator's certified weighbridge total (W_{ticket}) proportionally to the physical weight measured leaving each compactor:

$$W_{\text{allocated, A}} = W_{\text{ticket}} \times \frac{m_{\text{A}}}{m_{\text{A}} + m_{\text{B}}} \quad \text{and} \quad W_{\text{allocated, B}} = W_{\text{ticket}} \times \frac{m_{\text{B}}}{m_{\text{A}} + m_{\text{B}}}$$

  - Crucially, the sum of $W_{\text{allocated, A}} + W_{\text{allocated, B}}$ still equals **100% of the operator's certified weighbridge ticket** (W_{ticket}), so the operator's total revenue is unchanged while site-level cost accounting becomes fair (for example, a `4,460 kg` combined load split `2,230 / 2,230 kg` is corrected by `±1,267 kg` between the two paired sites).
- **Value to Customer & Europress:** Prevents customers from raising false-positive invoice disputes on combined loads (`271` trap cases avoided) and creates a powerful **fleet-wide network effect for Europress**: multi-site retail and logistics customers must equip *all* their sites with Europress SMART weighing so shared collection routes can be proportionally reconciled.

Appendix Idea 3: Verifying Compaction Force & Upgrading Overworked COMBI Units to DUO

- **Why It Matters:** Customers pay a fixed lift and transport fee (`€110–€350`) every time a truck collects a container. A fill sensor tells you *when* a container reaches `80%` volume, but it cannot tell you whether increasing hydraulic press force actually packed more kilograms into each cubic meter (`kg/m³`), or whether a site's annual weight throughput (`tonnes/year`) has outgrown a smaller **COMBI** unit (`<= 22 m³`), forcing the customer to pay for `45–152` truck visits per year.
- **How It Works:**
  - **Compaction Tuning Verification:** Compares measured compacted density (`kg/m³`) before and after hydraulic pressure adjustments (`16` compactors in the competition dataset) to prove the exact payload weight gain per lift.
  - **COMBI-to-DUO Right-Sizer (Respecting Machine-Class Constraints):** Scans all **non-biowaste COMBI** units (`<= 22 m³`, since biowaste must always remain in sealed one-piece **COMBI** containers) with high annual lift frequency and verified annual weight throughput. Recommends upgrading to a `24–30 m³` stationary **Europress DUO** compactor where the larger detachable container cuts annual truck visits by `35–50%`.
- **Value to Customer & Europress:** Saves the customer `€2,500–€4,200/yr` in fixed lift and transport fees while directly expanding **Europress's core equipment leasing business** (`38` upgrade candidates × `+€110/mo` incremental **DUO** lease = **+€50,160/yr recurring hardware lease revenue**).

Appendix Idea 4: Separating Truck Tare Errors from Compactor Sensor Drift

- **Why It Matters:** When a compactor's weight reading disagrees with an operator's weighbridge ticket, a naive single-pickup comparison cannot tell **whose scale is wrong**: *Did the compactor's load cells drift, or did the hauler's truck use an outdated pre-saved tare weight instead of weighing the empty truck?*
- **How It Works:**
  - Cross-checks **collection trucks** and **compactors** across clean single-stop pickups (with no loose side waste), verified against maintenance calibration logs.
  - **Truck Stored-Tare Isolation:** When a specific collection truck using a pre-saved tare weight consistently reports `+240 to +440 kg` more than multiple independent compactors across different sites (and diverges from its own live-weighed tare tickets), the discrepancy is isolated to the vehicle's stored tare table (`18` trucks across `1,531` pickups = `+260.9 tonnes` / `€12,206` in the competition dataset).
  - **Compactor Sensor Calibration Isolation:** When a single compactor diverges across multiple live-weighed trucks or coincides with a sensor recalibration or load-cell replacement record (`63` episodes in the competition dataset), the system suppresses operator invoice claims and triggers a **Europress technician calibration check**.
- **Value to Customer & Europress:** Achieves **100% neutral operator tone** (zero false accusations against waste operators when a compactor sensor drifts) and creates a high-margin **billable field calibration service (€185/visit)** for Europress.

Appendix Idea 5: Instant Pickup Weight Receipts & Unlocking Recyclable Credits

- **Why It Matters:** Cost per tonne cannot be calculated when collection weight is unknown, and per-tonne billing discrepancies have zero recoverable value without independent weight evidence. Yet across the fleet, **26.6% of completed pickups (4,908 collections)** have **no weighbridge ticket shared** (one regional operator shares `0%`), **2,991 tickets** arrive **more than 7 days late** (up to `116.8` days), and **36 of 52 customers (69%)** receive **0% material credit** on recyclable streams because they lack an independent tonnage ledger.
- **How It Works:**
  - **Day-0 Weight Receipt & Contract Verification Clause:** Generates an immediate provisional weight receipt (`Net Collected Weight ± 3.3%`, tagged `Estimated`) the moment a pickup finishes. When monthly invoices arrive, single-load per-tonne charges diverging by more than the contractual verification tolerance are flagged for human-approved **weighbridge data requests** (when tickets are missing) or **evidence-backed invoice reviews** (`2,814` single-load gaps `>10%` worth `€86,923`).
  - **Strict Credit-Eligibility Rule:** Evaluates contracts exclusively on credit-eligible recyclable streams (**cardboard, plastic film, and mixed plastic only** – never paper, mixed waste, or biowaste, which are strictly chargeable) where the customer currently receives zero credit share, quantifying the annual credit value unlocked at peer-median credit terms (`Hypothetical`).
- **Value to Customer & Europress:** Eliminates weeks of financial blind spots (`4,908` unshared pickups) and provides the verified recyclable tonnage ledger needed to negotiate material credit shares (`€365,671/yr` identified across `36` contracts).

Appendix Idea 6: Ordering Pickups at Target Weight to Avoid Minimum-Charge Penalties

- **Why It Matters (Zero Bridging Dependency):** On per-tonne and hybrid minimum-charge contracts (`35%` of the fleet), the waste operator charges a contractual minimum fee per pickup (`€140–€260`). Even when the fill sensor is completely accurate, if the waste inside is light or bulky, or if a site manager places a precautionary Friday pickup order before a weekend, lifting the container before its **weight** reaches the contract's minimum-charge break-even threshold triggers a **minimum-charge top-up penalty**.
- **How It Works:**
  - Computes the exact contractual break-even weight ($W_{\text{break-even}}$) where variable haulage and processing charges equal the contract's minimum charge per pickup:

$$W_{\text{break-even}} = \max\left(0, \frac{C_{\min} - C_{\text{fixed}}}{R_{\text{haul}} + R_{\text{proc}}}\right)$$

  - Forecasts live hourly container weight over the hauler's regional lead time and holds non-urgent dispatches until container weight reaches target payload – **while strictly enforcing the 7-day biowaste hygiene limit, maximum collection intervals, and zero container overflow** (preventing any hygiene or overflow penalty traps).
- **Value to Customer & Europress:** Across the competition dataset, `2,560` pickups incur **€192,789.15 in minimum-charge top-up penalties**, and switching eligible units to target-weight dispatch saves **€303,040/yr** across `304` compactors (`Hypothetical` counterfactual).